Key Compass Manual

Oct 9, 2026 · @Dave King

What Key Compass is

Key Compass listens to music and shows its key, the chord playing now, and the notes being played. Its second page, Song Mapper, turns chords and genre progressions into a song map you can drag into Cubase as MIDI. It comes in two forms: a VST3 plug-in for Cubase or any VST3 host, and a standalone app that listens to your microphone or audio input.

It only listens. Your audio passes through unchanged, so you can leave it on any track.

Installing and updating

The first install takes about 10 minutes. You need Visual Studio Community with the "Desktop development with C++" workload, plus Git.

1. Right-click `KeyCompass.zip`, choose **Properties**, tick **Unblock** if it's there, then click OK.
2. Right-click the zip, choose **Extract All**, and extract it to a short path such as `C:\KeyCompass`.
3. Double-click **Build and Install.bat**.
   - If "Windows protected your PC" appears, click **More info**, then **Run anyway**.
   - When Windows asks to copy into Program Files, click **Yes**.
4. When it says **All done!**, press any key to close the window.

What	Where it goes
VST3 plug-in	`C:\Program Files\Common Files\VST3\Key Compass.vst3`
Standalone app and a spare copy of the plug-in	The `Output` folder inside the project
Desktop shortcut	Your desktop, labelled "Key Compass"

To update: close Cubase and the app first. Extract the new zip into the same folder and click **Replace**, then run the .bat again. Reusing the folder keeps the downloaded JUCE files and earlier build, so updates are quick.

A tour of the screen

The window is a pocket compass with its lid open. The lid shows what was found, and the compass shows it live.

The **SONG MAPPER** plaque at the top left opens the second page. The **MANUAL** plaque at the top right opens this manual as a PDF.


SONG MAPPER
MANUAL
Key Compass
OREGON
DETECTED KEY
F♯ MINOR
CONFIDENCE
99%
KEY FAMILY
F♯m | A | Bm | C♯m | D | E
CHORDS
C♯ → F♯m → Bm → C♯ → F♯m
C
G
D
A
E
B
F♯
C♯
G♯
D♯
A♯
F
LIVE CHORD
AUDIO
F♯
NOTE
MODE
TRACK
RESPONSE
BALANCED
TUNING A4
440.0 Hz
LOCK KEY
FLATS
CLEAR HISTORY

The lid

Part	What it shows
Detected key	The key, glowing blue once it's found. Shows "Listening..." while it decides, or "Play some music" when there's no sound.
"or A major" (under the key)	A close second choice. It's usually the relative major or minor, which uses the same notes with a different home note.
Confidence bar and %	How clearly the winning key beats the others. Blue means confident; orange or red means still unsure.
Key family	The six main chords of the key. The chord playing now glows blue.
Chords	Recent chords, newest on the right. A blue diamond marks a key change. In Mono mode this becomes Notes.

The compass

Part	What it shows
Letters around the edge	All 12 notes, arranged in the circle of fifths. Next-door letters are closely related keys.
Needle	Points at the key's home note. While searching, it wanders gently.
Gold wedge	The seven notes that belong to the key. Letters outside it are faded.
Blue glow on a letter	The home note of the detected key.
Dashed blue ring	The home note of the runner-up key.
Gold or red letter glow	A note sounding right now. Gold means it's in the key; red means it's outside it.

Part	What it shows
Live plate (centre)	The current chord and the notes in it, and whether the source is AUDIO or MIDI. In Mono mode it shows the note with its octave, a tuning meter and the frequency.

When you stop playback, the last reading stays on screen in a dimmed state so you can still read it.

Controls

Every control is saved with your Cubase project and can be automated. Hover over any control for a short explanation.

Control	What it does	How to use it
Mode	**Track**: one instrument, showing chords and key. **Mono**: a single voice, bass or lead, showing the note, octave and cents. **Song**: the whole mix.	Click and pick from the menu.
Response	How quickly the key reacts. **Fast** follows changes quickly, **Balanced** is the default, and **Steady** waits for more music before deciding.	Click and pick from the menu.
Tuning A4	The reference pitch, normally 440 Hz (range 415–466 Hz).	Click the arrows for 1 Hz steps (hold Shift for 0.1 Hz), drag up or down, or scroll. Double-click resets it to 440.
Lock Key	Freezes the current key. Chords and notes keep updating. A locked key is saved with the project.	Click; it glows blue when on.
Flats	Shows flats (B♭, E♭) instead of sharps (A♯, D♯).	Click; it glows blue when on.
Clear History	Clears the chord list and starts listening for the key again. If Lock Key is on, the key is kept.	Click.

Control	What it does	How to use it
Window size	Makes the window bigger or smaller.	Drag the bottom-right corner, or right-click the background for Small, Medium, Large, Extra large or Full.

Using it in Cubase

Add Key Compass as an insert effect. It's listed under **Analyzer**. If it doesn't appear, open **Studio > VST Plug-in Manager** and click **Rescan**.

You want to know	Put Key Compass on	Mode
The chords and key of one part (guitar, piano, synth, a sample or loop)	That track, as an insert	Track
Which note a singer or bass is hitting, and how in tune it is	The vocal or bass track	Mono
The key of the whole song	The Stereo Out (master)	Song
Exact chords from a MIDI part	Any audio track, then route MIDI to it (below)	Track

Routing MIDI to it

MIDI gives exact notes instead of estimating them from sound.

1. Insert Key Compass on any audio or group track.
2. On your MIDI or instrument track, open **Output Routing** and choose **Key Compass**.
3. Press Play. The live plate switches from AUDIO to MIDI by itself.

Sustain pedal is supported, so held notes count while the pedal is down.

Tips

- Press **Clear History** when you move to a different song or section, so old chords don't sway the key.
- Once the key is clear, turn on **Lock Key** while you write. The chord readout and note glows keep working against that key.

- For a song that changes key, set Response to **Fast**. Key changes appear as blue diamonds in the Chords list.
- Key Compass adds no latency and uses very little CPU, so it's fine to leave it on several tracks.

The standalone app

The app works without Cubase and listens to your microphone or audio interface, so you can sing or play into it. Open it from the **Key Compass** shortcut on your desktop, or run `Output\Key Compass.exe`.

1. Click **Options** at the top of the window.
2. Under **Input**, choose your microphone or audio interface.
3. To use a MIDI keyboard, tick it under **Active MIDI inputs**.
4. Close the Options box and start playing or singing.

The app never plays your microphone back out, so there's no feedback squeal. Mono mode suits singing; Track mode suits a guitar or piano held near the mic.

Sharing Key Compass with a friend

Don't send your KeyCompass folder: building it needs Visual Studio. Instead, double-click **Make Shareable Package.bat** in that folder (after **Build and Install.bat** has run). It makes **Key Compass Shareable.zip** with the finished plug-in and app, this manual and a one-click installer. If you have changed anything since the last build, it offers to build first.

1. Send the zip with Google Drive, Dropbox, OneDrive or WeTransfer. Email often blocks zips with an app inside.
2. Your friend right-clicks the zip, chooses **Properties** and ticks **Unblock** if it is there, then extracts it.
3. They double-click **Install Key Compass.bat**. If Windows says "Windows protected your PC", they click **More info**, then **Run anyway**, and click **Yes** when it asks to change Program Files.

The plug-in goes into the VST3 folder and the app gets Desktop and Start menu shortcuts. **Uninstall Key Compass.bat** removes it. It works on 64-bit Windows 10 and 11.

Song Mapper

Song Mapper is the second page of Key Compass: pick chords or genre progressions, see how to play them, arrange them into song sections, and drag the finished song into Cubase as MIDI chords. Open it with the **SONG MAPPER** plaque at the top left of the compass; **KEY COMPASS** takes you back. The window grows for this page, and each page remembers its own size.


◂ KEY COMPASS
TAP TEMPO
CAPTURE CHORDS
SONG ▾
Song Mapper
?
DRAG SONG TO DAW
32 bars · 1:04
VOL
KEY
F♯ MINOR
heard - click to change
TEMPO
120 BPM
drag or tap · 4/4
CHORD DIAGRAMS
GUITAR
PIANO
STANDARD
F♯m (F♯ minor)
PROGRESSION
drag to reorder
F♯m
D
A
E
add
Type a chord...
KEY CHORDS
▶ PLAY
★ SAVE
CLEAR
ADD ALL TO "VERSE 1"
SONG STRUCTURE & CHORD TIMELINE
▶ PLAY SONG
■ STOP
LOOP
LINES
Intro
4 BARS
1 BAR EACH
F♯m
E
+ bass line
+ lead line
Verse 1
8 BARS
1 BAR EACH
F♯m
D
A
E
F♯m
D
Bm
C♯
Pre-Chorus 1
4 BARS
SPREAD OUT
Bm
D
E
Chorus 1
8 BARS
1 BAR EACH
D
A
E
F♯m
CHORDS
BASS
LEAD
PROGRESSION BROWSER
ROCK
ALTERNATIVE
Chords shown in f♯ minor
Open Drone
vi7 – IV – I – Vsus4
F♯m7 · D · A · Esus4
USE
Moody
i – ♭VI – ♭III – ♭VII
F♯m · D · A · E
USE
Major-to-Minor iv
I – III – IV – iv
A · C♯ · D · Dm
USE
Borrowed
I – ♭VII – IV – iv
A · G · D · Dm
USE
Strummed
PLAY STYLE
BLOCK
TIME SIG.
4/4
OCTAVE
NORMAL
LOCK KEY
FLATS
CLEAR SONG

The panels

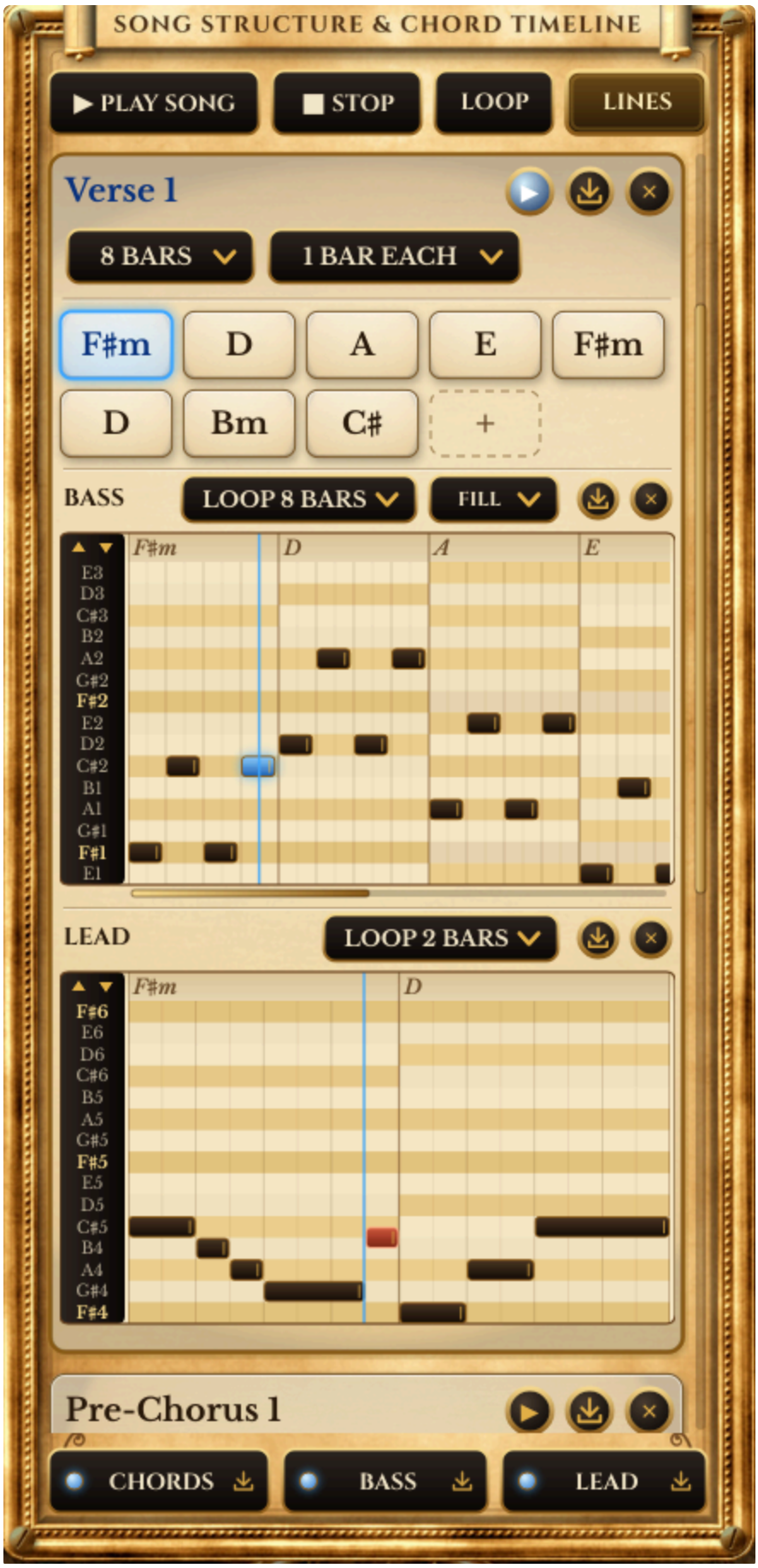
Panel	Where	What it does
Key and tempo	Top centre	The key follows what Key Compass hears or, when nothing is coming in, the key your chords are in (the line under it says which); click it to choose a key yourself. The tempo follows Cubase; drag or scroll it, or use **TAP TEMPO**. Click the line under the tempo for 4/4, 3/4 or 6/8.
Chord diagrams	Left	Shows the chord you last clicked, on **GUITAR** or **PIANO**. **VOICING** picks the shape: Standard, Open/Easy, Triad, Barre, Power Chord, Shell, Inversion or Color/Extended. Click a dot, an open string or a piano key to hear just that note.
Chord practice	Four-arrow button, top-right of the chord diagram	Opens the chords big, with finger numbers, so you can learn them by hand. See Practice chords with big diagrams below.
Progression	Left, under the diagram	Your working set of up to 8 chords. Type a chord (F♯m7, B♭maj9, D/F♯) and press Enter, use **KEY CHORDS**, or click the **+** card. Drag cards to reorder.
Progression browser	Right	Pick a **GENRE** and **STYLE**, or the star for your favourites. Every progression is shown in your key.
Song structure and chord timeline	Centre	Your song as sections (Intro, Verse 1, Chorus 1…), each with its chords, length and playback, plus optional bass and lead rows.

Panel	Where	What it does
Drag Song to DAW	Top right porthole	Drag it into Cubase to drop the whole song as MIDI. **VOL** under it sets the preview volume.

Under each section's name are two boxes: its **length** in bars (what it fills in your DAW) and **how long each chord lasts** (1 BAR EACH, ½ BAR EACH and so on). Chords repeat until the section is full; **SPREAD OUT** shares the bars evenly between them. **+ add a section**, at the bottom of the timeline, adds a new one. **PLAY SONG** plays every section that has something in it, so with **LOOP** on it goes straight back to the start of what you have written.

Bass and lead rows

Any section can have its own bass and lead line, written as notes in a small piano roll under its chords. Click **+ bass line** or **+ lead line** under a section's chords to add one; a new bass row starts from the chords' roots so you hear it straight away.


SONG STRUCTURE & CHORD TIMELINE
▶ PLAY SONG
STOP
LOOP
LINES
Verse 1
8 BARS
1 BAR EACH
F♯m
D
A
E
F♯m
D
Bm
C♯
+
BASS
LOOP 8 BARS
FILL
F♯m
D
A
E
E3
D3
C♯3
B2
A2
G♯2
F♯2
E2
D2
C♯2
B1
A1
G♯1
F♯1
E1
LEAD
LOOP 2 BARS
F♯m
D
F♯6
E6
D6
C♯6
B5
A5
G♯5
F♯5
E5
D5
C♯5
B4
A4
G♯4
F♯4
Pre-Chorus 1
CHORDS
BASS
LEAD

The rows of the roll are the notes of your key, plus any chord note from outside the key (its name is blue), so every note you place fits the chords. Gold rows are the notes of the chord playing above them, and the chord names run along the top.

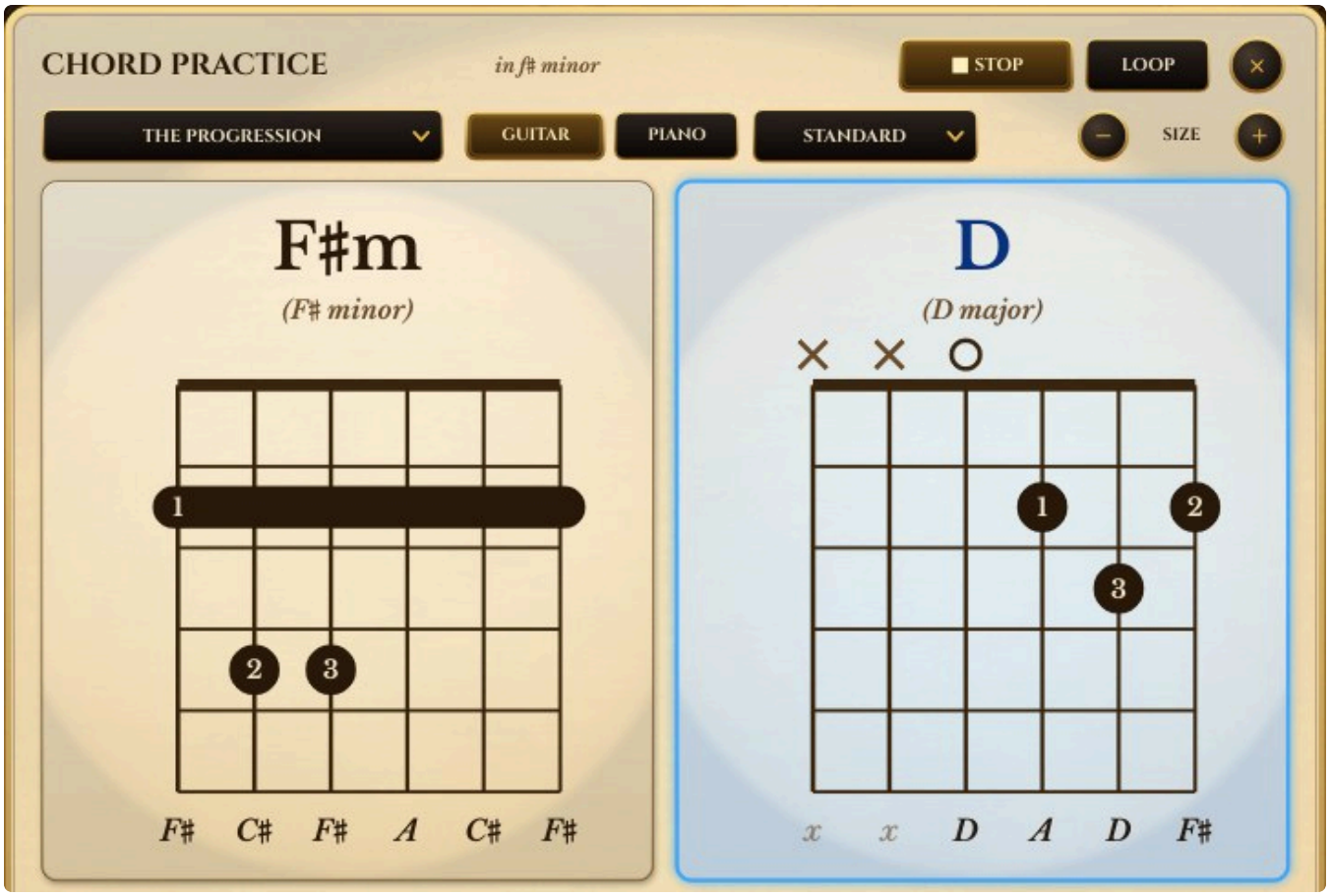
Do this	What happens
Click an empty spot	Adds a note there and plays it. Hold Alt for 16ths.
Drag a note	Moves it in time or up and down the key. Drag its right end to change its length.
Double-click a note	Deletes it (or select it and press Delete).
Up / Down	Moves the selected note to the next row up or down. Shift+Up/Down sharpens or flattens it; notes outside the key turn red.
Right-click a note	Play, length, sharpen, flatten, octave up or down, delete.
LOOP	How long the pattern is before it repeats: 1, 2, 4 or 8 bars, or the whole section.
FILL (bass only)	Writes a bass line from the chords: Roots (held), Root pulse (8ths), Root and fifth, Octaves or Walking. Then change any note you like.
Arrows above the note names	Move the rows up or down an octave.
Arrow button, ×	Drag just this row of this section into Cubase; hide the row (its notes are kept).

You can edit while the song plays and hear the change on the next pass. **LINES**, next to LOOP above the timeline, shows or hides every bass and lead row; while they are hidden only the chords are played and dragged. The three pills under the timeline, **CHORDS**, **BASS** and **LEAD**, mute that part when you play the song, and drag that part of the whole song into Cubase.

Practice chords with big diagrams

The small chord diagram is fine for a glance but too small to learn from. Click the round button with four arrows at the top-right of the diagram to open **CHORD PRACTICE**: the chords fill the window, each guitar dot is numbered with the finger to use (1 index, 2

middle, 3 ring, 4 little), and the note on each string is written underneath. You can also right-click a progression card and choose **Practice view**, or right-click a section name and choose **Practice these chords**.



Control	What it does
Top-left box	Which chords: the progression, one section of your song, or just the chord you picked
GUITAR / **PIANO** and the voicing box	The same as on the small diagram. On piano every chord shares one keyboard, so you can see your hand move from chord to chord
− SIZE +	Fewer, bigger diagrams per page, or more, smaller ones (the + and − keys do the same). Guitar and piano each remember their size
PLAY / **LOOP**	Plays the chords. The one sounding glows blue and the pages turn on their own, so you can play along

Control	What it does
Click	A chord to hear it, or a dot, open string or piano key to hear that one note. While the chords are playing, clicking a card just picks it, so it doesn't sound over the music
Drag a card	Moves that chord to a new place. This rearranges the progression or the song section itself, and if they are playing they carry on in the new order. Hold the card over ◀ or ▶ to go to another page. You can also click a card and press Shift+← or Shift+→, or right-click it and choose a Move option. Ctrl+Z undoes a move.
◀ ▶ or the arrow keys	Turn the page when there are more chords than fit
× or Esc	Close

The controls at the bottom

On the Song Mapper page, the six brass plates at the bottom hold Song Mapper settings instead of detector settings. The compass keeps its own Mode, Response and Tuning, and they come back when you switch pages.

Plate	What it does
Play Style	How chords are played, both in the preview and in the MIDI you drag into Cubase. **Block** holds each chord, **Pulse** plays straight 8ths, **Strum** plays a down-up guitar strum, **Arpeggio** plays one note at a time.
Time Sig.	4/4, 3/4 or 6/8. It sets how long a bar is and is written into the MIDI.
Octave	Moves the chords up or down by up to two octaves, in the preview and in the MIDI. The bass and lead rows keep their own notes. Double-click for Normal.

Plate	What it does
Lock Key	Keeps the Song Mapper in the current key instead of following what Key Compass hears or the key of your chords. Turn it off to follow again; with nothing heard and no chords yet, it keeps the last key you set. It lights up when you pick a key yourself.
Flats	Same as on the compass: spells chords with flats.
Clear Song	Removes the chords and the bass and lead notes from every section but keeps the sections. Ctrl+Z brings them back.

Map a song, step by step

1. Play some of your song so the key appears, or click the key and choose one.
2. Click **SONG ▾** > **New song from a template** (Pop, Rock, Hip-hop, EDM, Worship or Country), or build sections with **+ add a section**.
3. In the browser, pick a genre and style and press ▶ to preview progressions. Press **USE** to load one into the progression on the left, or drag the row straight onto a section.
4. Click any chord to hear it and see its shape. Change the voicing until it suits your playing.
5. Click a section to select it, then press **ADD ALL TO "…"**, or drag single chords from the progression onto any section.
6. Set each section's length and how long each chord lasts, then press **PLAY SONG** to hear it in order (empty sections are skipped). Drag sections by their name to rearrange them.
7. For a bass or lead line, click **+ bass line** or **+ lead line** under a section's chords and shape it in the piano roll.

Dropping it into Cubase

1. Drag the **DRAG SONG TO DAW** porthole onto a MIDI or instrument track, or below the last track to make a new one.
2. The song lands as MIDI chords with the section lengths you set, played in the Play Style you chose, plus a marker for each section.

3. If the song has bass or lead rows, the file holds one MIDI track per part (chords, bass, lead), so Cubase can make a track for each. To put one part on a track you already have, drag its pill (**CHORDS**, **BASS** or **LEAD**) under the timeline, or the arrow button on that row.
4. To drop one section, drag the arrow button on that section's row instead.
5. Click the porthole to save a .mid file if you would rather import it.

Tips

- Right-click chords, sections and browser rows for more options (change chord, duplicate, rename, move, add as a new section).
- Double-click a section name to rename it, or a browser row to add it to the selected section.
- **SONG ▾** has undo and redo; **Ctrl+Z** and **Ctrl+Shift+Z** work too after clicking in the page.
- **CAPTURE CHORDS** copies the chords Key Compass just heard into the progression.
- The star on a browser row, or **★ SAVE** under the progression, adds it to your favourites. Favourites are shared by every project.
- The preview sound plays through the track Key Compass sits on (in the standalone app, through your speakers). Your song is saved with the Cubase project.
- In flat keys (F, B♭, Dm…) chords are spelled with flats automatically; the **FLATS** button forces flats everywhere.
- The preview uses recorded instruments: an acoustic guitar (or a piano, with **PIANO** selected) for the chords and lead, and an electric bass for the bass rows.
- This manual is built in: **MANUAL** at the top right of the compass, or the **?** button beside the porthole in Song Mapper (or press F1). It opens as a PDF in your PDF viewer or web browser.

Understanding the results

The key needs a few bars of music before it settles. Confidence starts low and rises as Key Compass hears more.

Confidence measures how far the winning key is ahead of the others, not whether the answer is right. A low number with a runner-up shown means the music genuinely fits two keys.

Relative major and minor are the most common close call. F♯ minor and A major use exactly the same seven notes. A loop such as F♯m–D–A–E fits both equally, so you'll see something like "F♯ MINOR 31% · or A major" with an orange bar. The deciding factor is which chord feels like home. Usually that's the chord the song starts or ends on, or the one the bass leans on most.

Red letters are notes outside the key. Often that's a borrowed chord, a passing note or a wrong note. In a minor key, the raised seventh (E♯ in F♯ minor) counts as part of the key, because songs use it so often.

Slash chords like D/F♯ appear when the bass plays a chord note other than the root.

How accurate each source is

Source	Notes and chords	Key
MIDI	Exact	Very reliable
One instrument (Track)	Reliable when the playing is clear	Reliable after a few bars
Single voice or bass (Mono)	Note and octave reliable, tuning within about ±5 cents	Reliable once the melody has used most of the scale
Full mix (Song)	Live chord names are rougher, because drums and bass blur them	Solid after several bars

Octaves (like C♯4) are shown only in Mono mode and from MIDI. When several notes sound together from audio, the octave can't be told apart.

Troubleshooting and FAQ

Problem	Fix
The build failed	Open `build-log.txt` in the project folder. The error is near the bottom.
"Visual Studio is installed, but without the C++ tools"	Open **Visual Studio Installer**, click **Modify**, tick **Desktop development with C++**, then run the .bat again.

Problem	Fix
"Could not copy the plug-in"	Close Cubase, then run the .bat again.
Key Compass isn't in Cubase	**Studio > VST Plug-in Manager > Rescan**. Check that `Key Compass.vst3` is in `C:\Program Files\Common Files\VST3`.
Nothing happens when I press Play	Make sure the track has audio coming through it. Key Compass needs sound above a very low level.
The key keeps changing	Set Response to **Steady**, or turn on **Lock Key**.
The key is the relative major or minor of what I expected	Both keys share the same notes. Trust your ear for the home chord, then use **Lock Key**.
Every note looks slightly sharp or flat	The recording may not be tuned to 440 Hz. Adjust **Tuning A4** until the tuning meter centres.
The desktop icon still looks blank	Right-click the desktop and choose **Refresh**, or sign out and back in.
The app hears nothing	Click **Options** and choose the right input device.
I want to start completely fresh	Delete the `build` folder and run the .bat again.
No preview sound in Song Mapper	The sound comes out of the track Key Compass is on. Make sure that track isn't muted and its fader is up, and VOL under the porthole isn't at zero.
Dragging the song into Cubase does nothing	Drop it on a MIDI or instrument track, or below the last track. Or click the porthole, save the .mid file and import it.

Problem	Fix
The Song Mapper chords are in the wrong key	Click the key at the top of Song Mapper and choose the key you want. Choose "Follow the detected key" to go back to automatic: it uses what Key Compass hears, or the key of your chords when nothing is coming in. Key Compass never hears the Song Mapper's own playback.
The bass or lead sounds out of tune against the chords	The key at the top probably doesn't match the chords, so notes were placed a half step off (F under an F♯m chord, say). Turn Lock Key off so the key follows your chords, or pick the right key, then move any red notes onto the gold rows.
PLAY SONG skips a section	Sections with no chords or notes are skipped while you preview, so a loop never plays bars of silence. They still count in the MIDI you drag into Cubase.
The standalone app updated but Cubase still shows the old plug-in	Cubase was open, so the old plug-in file couldn't be replaced. Close Cubase completely and run **Build and Install.bat** again, clicking **Yes** when Windows asks. Or copy `Output\Key Compass.vst3` into `C:\Program Files\Common Files\VST3`, replacing the old one. Right-click the background of Key Compass to see the version and build date you're running.

Credits

Guitar and bass sounds: recordings from tonejs-instruments by Nicholaus P. Brosowsky, licensed CC BY 3.0. The acoustic guitar was recorded by the University of Iowa Electronic Music Studios and the electric bass by Karoryfer Samples; both were trimmed and re-encoded for Key Compass. Built with the JUCE framework. Fonts: Cinzel, Libre Baskerville and Pirata One (SIL Open Font License).